Tab 1

Software Requirements Specification

for

Money Manager System

Version 1.0 approved

Prepared by Group 5

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Revision History

Name	Date	Reason For Changes	Version
Group 5	01/03/2026	Filled out Chapter 1 of the SRS	1.0
Group 6	16/03/2026	Chapter 2 Progress	1.1

		Filled out chapter 2.1 and initial progress for UML diagram, use case diagram, activity diagram, and figma design.	
Group 6	01/04/2026	Chapter 2 Progress Filled out chapter 2.3 and made UML diagram, use case diagram, activity diagram, and figma design.	1.2
Group 6	14/04/2026	Chapter 2 Progress Filled out chapter 2.3 - 2.7 and finalizing UML diagram, use case diagram, and activity diagram.	1.3
Group 6	16/04/2026	Chapter 3 Progress Finalizing figma design and filled out chapter 3 of the SRS	1.4
Group 6	21/04/2026	Filled out Chapter 4 of the SRS	1.5
Group 6	23/04/2026	Filled out Chapter 5 of the SRS	1.6

1. Introduction

1.1 Purpose

This document describes the software requirements for Website Money Manager, a web-based personal finance management application. It is intended to guide the development team in building and delivering the system according to the defined requirements.

1.2 Document Conventions

<Describe any standards or typographical conventions that were followed when writing this SRS, such as fonts or highlighting that have special significance. For example, state whether priorities for higher-level requirements are assumed to be inherited by detailed requirements, or whether every requirement statement is to have its own priority.>

- Skip??

1.3 Intended Audience and Reading Suggestions

This document is intended for the following audiences:

a. *Developers*

   Responsible for building and implementing the system. They should read the entire document, with focus on:

   Section 3 – Functional Requirements (what features to build)
   - *Section 4 – External Interface Requirements (how the UI/system should behave)*
   - *Section 5 – Non-Functional Requirements (performance, security, etc.)*

b. *Project Manager*

   Responsible for planning, tracking progress, and ensuring the project stays on scope. They should focus on:

   - *Section 1 – Introduction (project overview and goals)*
   - *Section 2 – Overall Description (system scope and constraints)*
   - *Section 3 – Functional Requirements (feature checklist for planning)*

c. *Project Owner*

   The decision-maker who defines the vision of the product. They should focus on:

   - *Section 1 – Introduction (to confirm the purpose aligns with their vision.*
   - *Section 2 – Overall Description (product scope and assumptions)*
   - *Section 3 – Functional Requirements (to approve the feature list)*

1.4 Product Scope

The Website Money Manager is a web-based personal finance management application that allows users to record, track, and analyze their financial transactions in an organized and structured way.

No	*Feature*	*Description*
1	*User Authentication*	*Users can register and log in to keep their data safe and consistent.*
2	*Transaction Input*	*Users can add income and expense transactions with category labels.*
3	*Wallet View*	*Users can see current balance along with visual analytics.*
4	*Transaction History*	*Users can view past transactions filtered by category and time period.*
5	*Multiple Transaction Books*	*One user can create and manage multiple transaction books.*

1.5 References

No external references were used in the preparation of this document, other than IEEE Std 830-1998 as the SRS template standard.

2. Overall Description

2.1 Product Perspective

Website Money Manager is a web-based personal finance management application designed to help users record, manage, and analyze financial transactions in a structured and organized way. The system is intended to function as an independent web platform that can be accessed through a browser without requiring any additional software installation.

This product is developed to simplify the management of daily income and expenses, enabling users to monitor their financial condition in real time. The application includes a user authentication feature to ensure data security and consistency, a transaction input feature for recording income *and expense transactions with category labels, and a wallet view feature that displays the current balance along with visual analytics.*

In addition, the system provides a transaction history feature that allows users to review past transactions filtered by category and time period. To support more flexible financial management, the application also allows a single user to create and manage multiple transaction books, which can be used for different purposes such as personal finances, business records, or savings plans.

Overall, Website Money Manager is positioned as an integrated personal finance information system that is user friendly, secure, and efficient in supporting users' financial recording and evaluation activities.

2.2 Product Functions

The main functions of the Website Money Manager are to help users manage their personal financial data efficiently, securely, and systematically. The system provides the following core functions:

1. User Authentication
The system allows users to register, log in, and securely access their personal accounts. This function ensures that each user's financial data is protected and only accessible by authorized users.

2. Transaction Input
Users can record financial transactions by entering income or expense data. Each transaction can be categorized using labels, making it easier to organize and track financial activities.

3. Wallet View
The system provides a wallet dashboard that shows the user's current balance. It also presents basic visual analytics so users can quickly understand their financial condition.

4. Transaction History
Users can view and review their previous transactions through a transaction history page. The system supports filtering transactions by category and time period to make searching and analysis more convenient.

5. Multiple Transaction Books Management
The system enables a user to create and manage multiple transaction books. This function allows users to separate financial records for different purposes, such as personal use, business activities, or specific budgeting plans.

6. Financial Monitoring and Analysis
The system helps users monitor their financial activities by summarizing recorded transactions and presenting simple analytical insights. This function supports better financial planning and decision-making.

2.3 User Classes and Characteristics

The system identifies two primary user classes:

Guest

Guest users are individuals who access the Money Manager System without being authenticated. They have limited access to the system and can only interact with the Register and Login features. Their role is to create a new account or sign in to an existing account before accessing the main financial management functions of the application. Guests are expected to have basic digital literacy, such as the ability to use a web browser, fill out online forms, and navigate simple web pages.

User

Users are registered and authenticated individuals who access the main features of the Money Manager System. They interact with functions such as managing transaction books, adding transactions, viewing transactions, checking wallet balance, and viewing charts or analytics. Users can create, edit, delete, and select transaction books, as well as record income and expenses, assign categories, and add notes to transactions. They are expected to have basic digital literacy and a general understanding of personal financial recording. Users rely on the system to organize and monitor their financial data efficiently and securely.

2.4 Operating Environment

Money Manager System will operate as a responsive web application, accessible through modern web browsers on both desktop and mobile devices. The system is designed to help users manage personal financial records conveniently anytime and anywhere, especially for recording transactions and monitoring balances through smartphones or laptops.

Component	*Specification*
Client Interface	*Responsive web application (desktop & mobile browsers)*
Supported Browsers	*Latest 2 versions of Google Chrome, Mozilla Firefox, Safari (iOS)*
Backend Runtime	*Node.js*
Database	*Relational database (MySQL 8.0+ or PostgreSQL 13+)*
Hosting (prototype)	Local machine (XAMPP / Docker) or free-tier cloud (e.g., Railway, Supabase)

2.5 Design and Implementation Constraints

The design and implementation of the Website Money Manager are subject to several constraints based on the scope of the prototype and the selected system design.

- *Web-Based Scope: The system is implemented as a web-based application. Native mobile application development for Android or iOS is outside the scope of this prototype.*

- *Responsive User Interface: The system interface must support responsive design so that it can be accessed properly on both desktop and mobile browsers. The layout should adapt to different screen sizes while maintaining usability.*
- *Personal Finance Scope: The system is intended for individual users to manage personal financial records. Shared accounts, collaborative financial management, or multi-user bookkeeping are not included in this prototype.*
- *Manual Financial Input: Financial data such as income, expenses, account information, and transaction categories are entered manually by the user. Integration with banking systems, e-wallets, or automatic transaction import is not included.*
- *Basic Analytics Only: The system provides only basic charts, wallet summaries, and transaction statistics based on recorded data. Advanced analytics, financial prediction, and AI-based recommendations are outside the scope of the current implementation.*
- *Book-Based Organization: The system supports multiple financial books, but each book is managed separately. Automatic synchronization or dependency between books is not included.*
- *Technology Constraint: The system must be implemented using web technologies, with Node.js as the backend runtime and a relational database such as MySQL or PostgreSQL for data storage.*
- *Prototype Deployment Limitation: The prototype is intended to run on a local machine or free-tier cloud hosting. Therefore, scalability, storage, and performance may be limited compared to a full production environment.*
- *UI Design Consistency: The user interface should follow a consistent visual design across modules such as Login, Register, Books, Wallet, Charts, and Profile pages. Navigation and feature placement should remain simple and easy to understand.*
- *Documentation Conventions: All high-level requirements should use unique IDs. User interface elements should be written in bold, and user roles should be written in italics throughout the technical documentation.*

2.6 User Documentation

The Website Money Manager project will deliver the following documentation components:

- *Software Requirements Specification (SRS): This document, detailing all functional and non-functional requirements of the system.*
- *User Interface Specification: A separate document or design prototype containing detailed user interface designs and screen layouts for the main features, including Login, Register, Books, Wallet, Charts, and Profile pages.*
- *User Guide: A guide explaining how users can register an account, log in, create and manage transaction books, record income and expenses, view wallet balance, review transaction history, and access charts or analytics.*

2.7 Assumptions and Dependencies

The following assumptions and external dependencies apply to the Website Money Manager system:

Assumptions:

- *All users are assumed to have a valid email address for registration and login purposes.*

- *Users are assumed to have access to a device with an internet connection and a modern web browser.*
- *Users are assumed to enter their financial data manually and correctly.*
- *Each user is responsible for managing their own transaction books and financial records.*
- *The system assumes that one account is used by one user for personal finance management.*

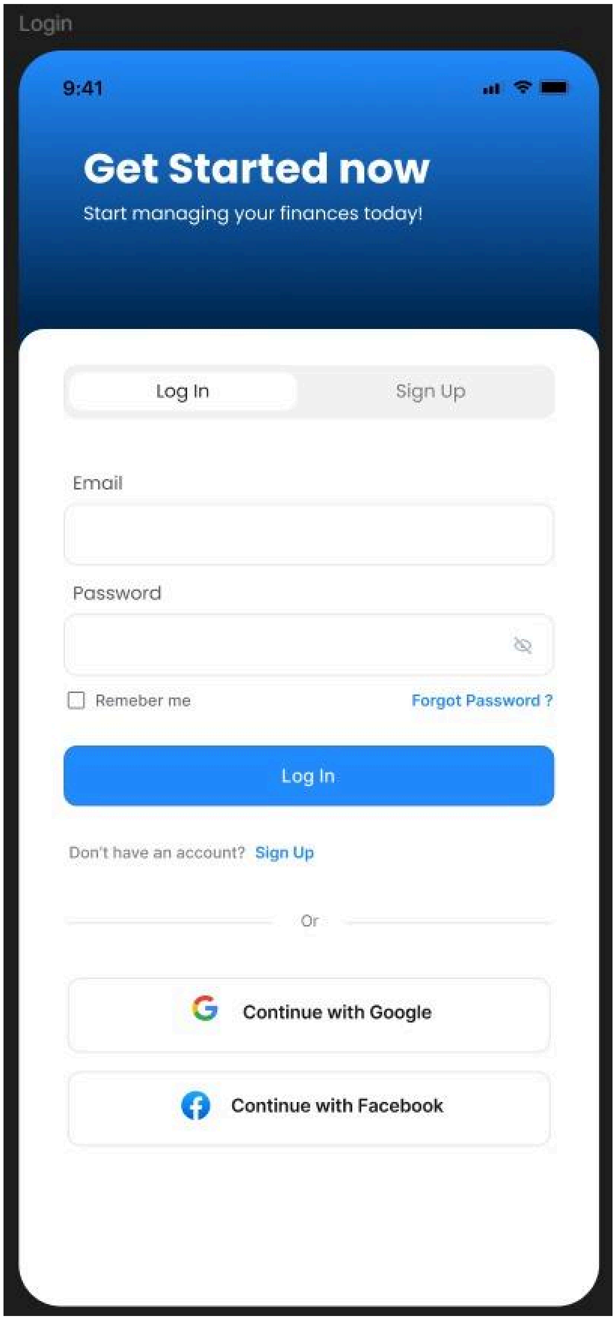
Dependencies:

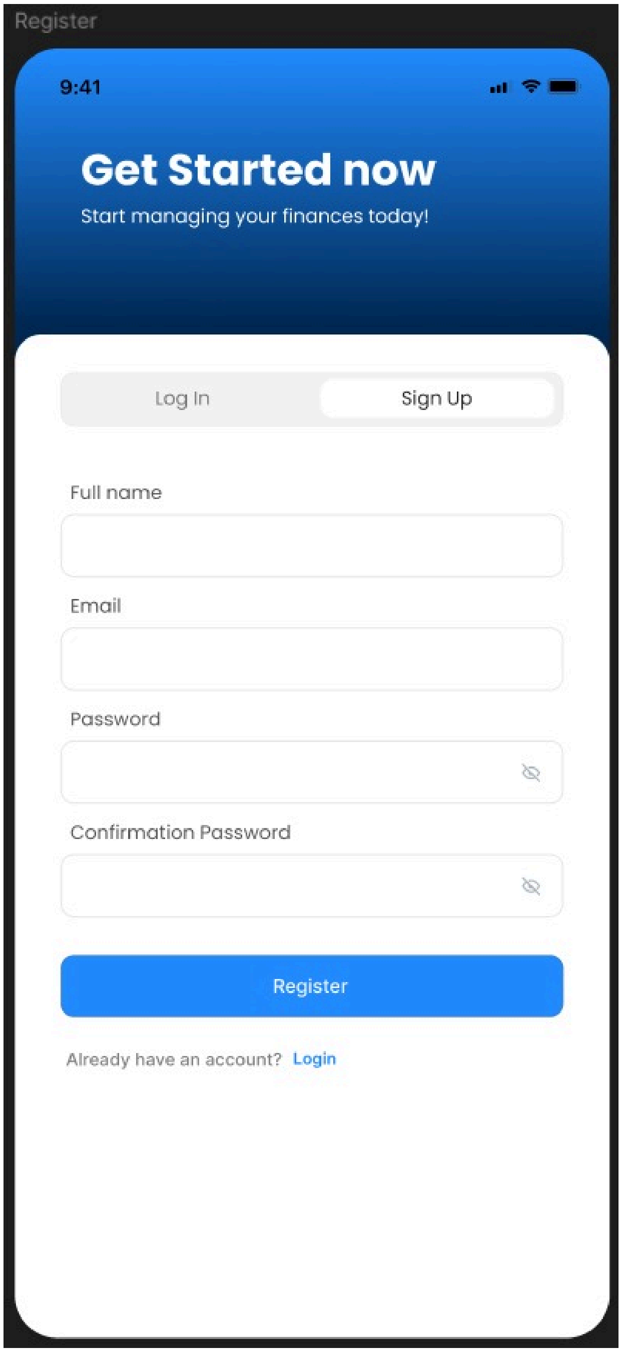
- *The system depends on a stable internet connection for accessing and updating financial data.*
- *The system depends on a backend server and database to store, process, and display user information and transactions.*
- *User authentication depends on the login and registration module working properly.*
- *Financial summaries, wallet balance, and charts depend on the transaction data entered by the user.*
- *The system depends on the selected technologies, such as Node.js and MySQL or PostgreSQL, for implementation and operation.*

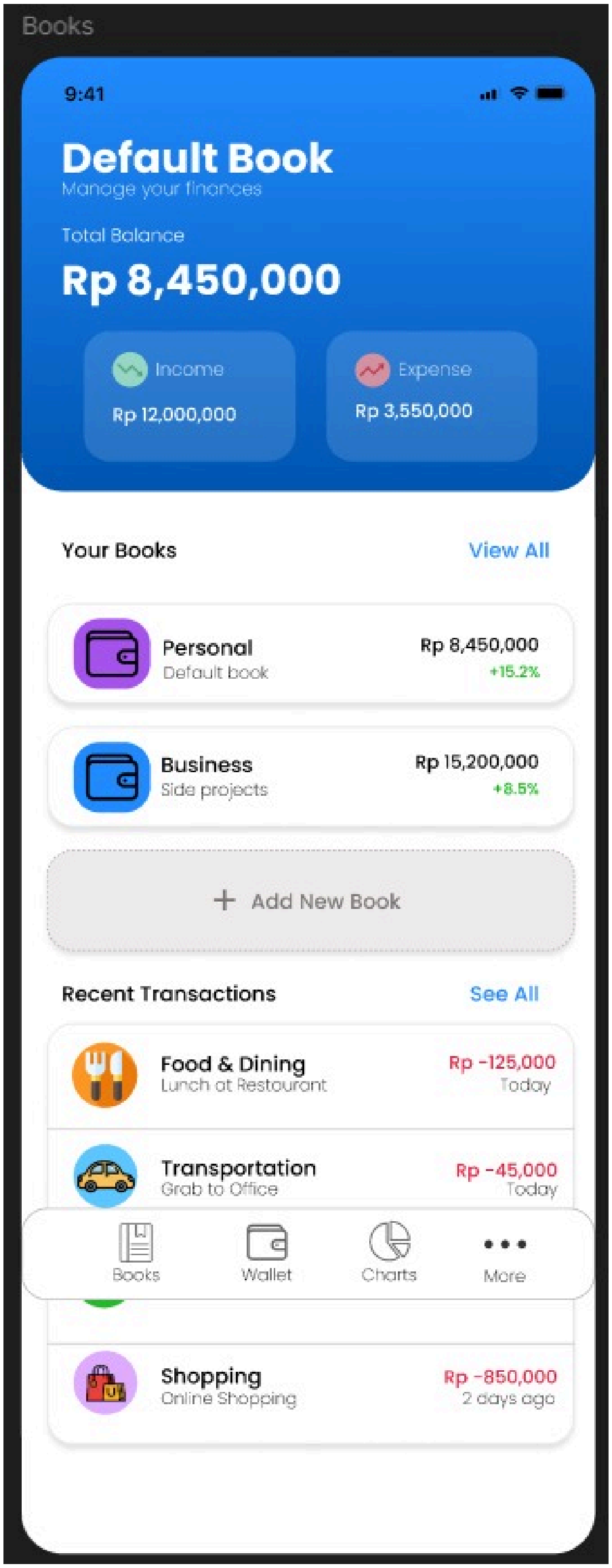
3. External Interface Requirements

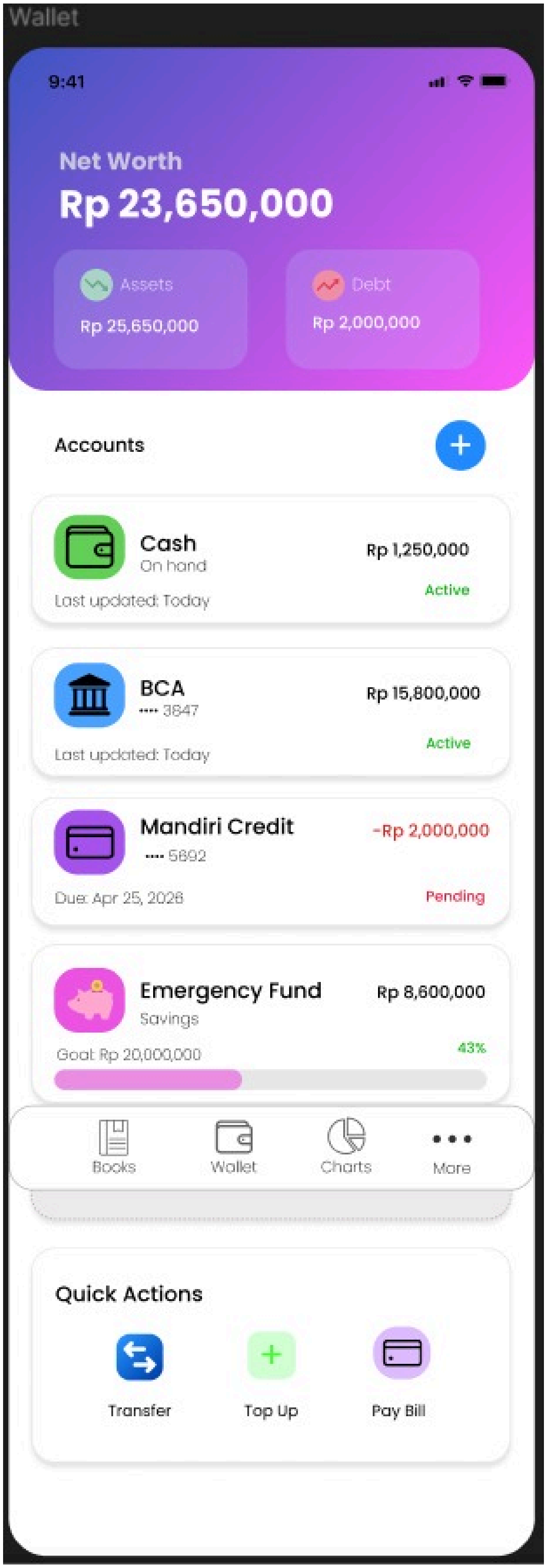
3.1 User Interfaces

Gambar Figma	keterangan

	The Login page provides an interface for existing users to access the system. Users are required to enter their email and password to authenticate. The page also includes a "Remember Me" option, a "Forgot Password" link, and alternative login methods such as Google and Facebook. A navigation option to the Sign Up page is also provided for new users.

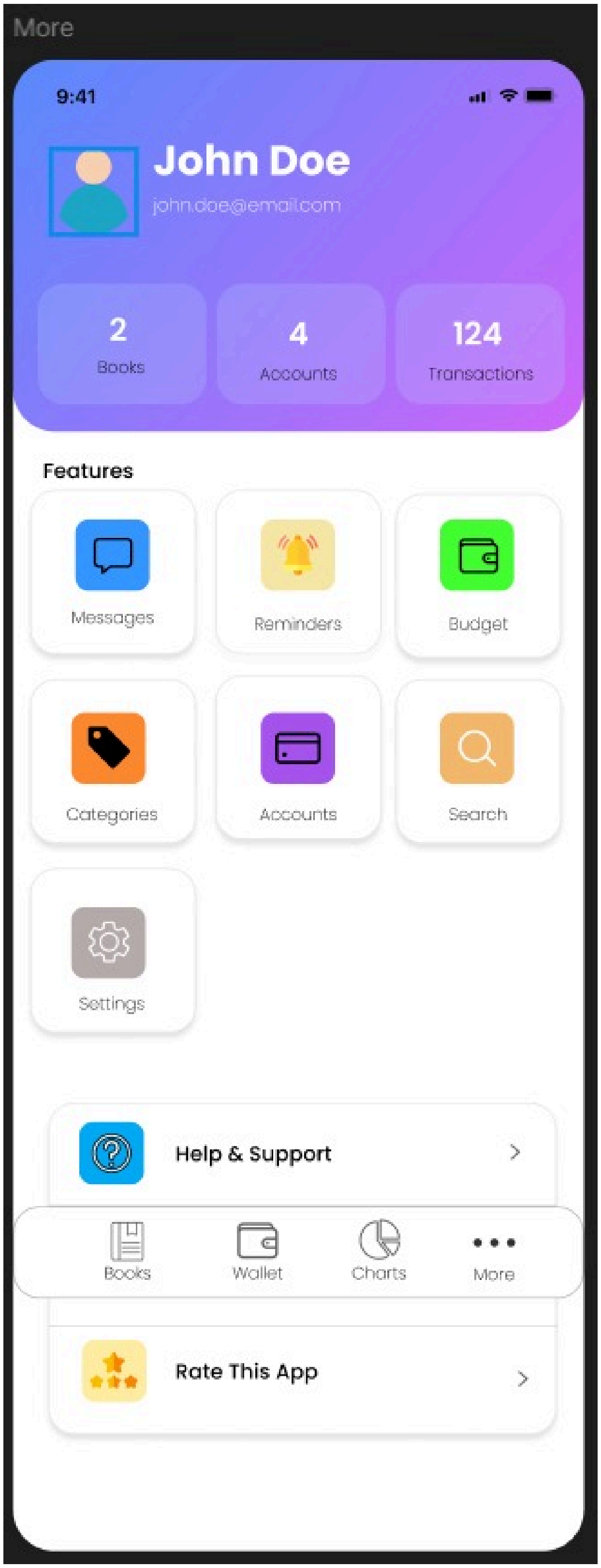
	The Register page allows new users to create an account by entering their full name, email, password, and password confirmation. The system validates user input before account creation. A navigation option is provided for users who already have an account to return to the Login page.

	The Books page serves as the main dashboard where users can view their financial overview. It displays the total balance, income, and expense summaries. Users can manage multiple transaction books, view recent transactions, and add new books. Navigation to other main features such as Wallet, Charts, and More is provided through a bottom navigation bar.



The Wallet page displays the user's net worth along with asset and debt summaries. It provides a list of user accounts such as cash, bank accounts, and savings. Users can perform quick actions such as transfer, top-up, and bill payment. The page also supports account status indicators such as active and pending.

The Charts page provides visual analytics of the user’s financial data. It includes expense summaries, category-based charts, and monthly trends. Users can switch between different time ranges such as monthly and yearly views to analyze their financial activity.

<table>
<tr>
<td>
</td>
<td>The Profile (More) page displays user information such as name and email, along with summary statistics including total books, accounts, and transactions. It provides access to additional features such as messages, reminders, budgeting tools, categories, account management, and settings. Support and feedback options are also available.</td>
</tr>
</table>

The Website Money Manager interface is designed to be simple, clear, and consistent across both desktop and mobile viewports. Because users may access the system frequently to record transactions or check their balance, the interface must support easy navigation, readable information display, and efficient data input.

- *GUI Standards and Product Style Guide:*
  - *Design Language: The interface shall follow a simple and modern web design style with clear page structure, card-based content grouping, and consistent spacing between elements.*
  - *Color Palette: Primary actions such as Login, Register, Add Transaction, and Save shall use the main system color. Negative or destructive actions such as Delete shall use a standard warning color such as red.*
  - *Typography: Sans-serif fonts shall be used throughout the system to ensure readability on different screen sizes.*

- *Screen Layout Constraints:*

  - *Responsive Design: All interfaces must be responsive across desktop and mobile devices, so users can access the system comfortably using different screen sizes.*

  - *Navigation: The system shall provide clear navigation between main pages such as Books, Wallet, Charts, and Profile.*

- *Standard Buttons and Functions:*
  - *Authentication Buttons: The Login and Register buttons shall be clearly displayed on the authentication pages.*
  - *Transaction Actions: The Add Transaction, Edit, Delete, and Save buttons shall be provided where needed for transaction and book management.*
  - *Book Management Actions: Users shall be able to create, select, edit, and delete transaction books through visible action controls.*
- *Error Message Display Standards:*
  - *All validation errors, warnings, and success messages shall be displayed clearly on the interface.*
  - *Success Messages: Displayed when an action such as login, registration, or saving a transaction is completed successfully.*
  - *Error Messages: Displayed when input data is incomplete, invalid, or when a system action fails.*
- *Software Components Requiring a UI:*
  - *Authentication Module: Includes Login and Register pages.*
  - *Books Page: Displays and manages available transaction books.*
  - *Wallet Page: Displays current balance and financial summary.*
  - *Transaction Page: Allows users to add and view income or expense transactions.*
  - *Charts Page: Displays simple financial charts and summaries.*
  - *Profile Page: Displays user account information.*

3.2 Hardware Interfaces

The Website Money Manager has minimal hardware interface requirements because it is a web-based application.

- *User Device: The system can be accessed using a smartphone, tablet, laptop, or desktop computer.*
- *Input Devices: Users interact with the system using standard input devices such as a keyboard, mouse, or touchscreen.*
- *Server Hardware: The system requires server hardware or cloud infrastructure to run the application and store financial data.*
- *Network Device: An internet connection is required so the user device can communicate with the server.*

3.3 Software Interfaces

The Website Money Manager interacts with several software components to support its operation.

- *Web Browser: The system is accessed through web browsers such as Google Chrome, Mozilla Firefox, Safari, or Microsoft Edge.*
- *Backend Framework: The system uses Node.js with a framework such as Express.js to handle application logic and client requests.*
- *Database Management System: The system uses a relational database such as MySQL or PostgreSQL to store user data, transaction books, and financial transactions.*
- *Operating System: The system can run on common operating systems such as Windows, macOS, and Linux on the server side, and Android or iOS on the client side through browsers.*
- *External Services: If needed, the system may connect to external services such as email services for account-related features like verification or password recovery.*

3.4 Communications Interfaces

- *Protocol: The system uses HTTPS for communication between the client and server to keep user data secure during transmission.*
- *Architecture: The system uses a RESTful API architecture. GET is used to retrieve data, POST to add new data, PUT/PATCH to update data, and DELETE to remove data.*
- *Data Format: Data exchanged between the client and server uses JSON format.*
- *Real-Time Update: Transaction, wallet, and book data are updated through normal client-server requests when the user performs an action or refreshes the page.*

4. System Features

This section describes the main features provided by the Website Money Manager system. Each feature explains the services offered to users and the main functional requirements related to that feature.

4.1 User Authentication and Registration

4.1.1 Description and Priority

Priority: High. This feature forms the entry point of the system. All core functionality, including transaction book management, transaction recording, wallet viewing, transaction history, charts, and profile access, is tied to an authenticated user *account. Registration is required before users can access the main features of the system.*

4.1.2 Stimulus/Response Sequences

- ***Stimulus:*** *A new user opens the Website Money Manager application and selects* ***Register****.*
- *System Response: The system displays a registration form requesting the user's name, email address, and password.*
- ***Stimulus:*** *The user submits the registration form with valid data.*
- *System Response: The system creates a new user account and redirects the user to the main page or login page.*
- ***Stimulus:*** *An existing user enters valid credentials on the Login page.*
- *System Response: The system validates the credentials and grants access to the user dashboard or main application page.*
- ***Stimulus:*** *A user submits invalid login credentials.*
- *System Response: The system displays an error message indicating that the email or password is incorrect and denies access.*
- ***Stimulus:*** *A user leaves one or more required fields empty during registration or login.*
- *System Response: The system displays a validation message requesting the user to complete the required fields before proceeding.*

4.1.3 Functional Requirements

- **REQ-4.1.1:** The system shall provide a registration feature for new users.
- **REQ-4.1.2:** The system shall require users to enter a name, email address, and password to create an account.
- **REQ-4.1.3:** The system shall validate that the email address is in a proper format before creating the account.
- **REQ-4.1.4:** The system shall store user passwords securely in the database.
- **REQ-4.1.5:** The system shall provide a login feature for registered users.
- **REQ-4.1.6:** The system shall require a valid email address and password for login.
- **REQ-4.1.7:** The system shall deny access if the entered login credentials are invalid.
- **REQ-4.1.8:** The system shall display an error message when login or registration fails.
- **REQ-4.1.9:** The system shall allow authenticated users to access the main features of the application.
- **REQ-4.1.10:** The system shall provide a logout feature for authenticated users.

4.2 Manage Transaction Books

4.2.1 Description and Priority

Priority: High. This feature is important because transaction books are used to organize the user's financial records. Through this feature, authenticated Users can *create, edit, delete, and select transaction books for different financial purposes, such as personal expenses, savings, or business records.*

4.2.2 Stimulus/Response Sequences

- ***Stimulus:*** *An authenticated User opens the Manage Transaction Books feature.*
- *System Response: The system fetches and displays all transaction books that belong to the user.*
- ***Stimulus:*** *User selects Create Book.*
- *System Response**:** The system displays a form for entering the new transaction book information.*
- ***Stimulus:*** *User submits valid book data.*
- *System Response: The system creates the new transaction book and updates the book list.*
- ***Stimulus:*** *User selects Edit Book on an existing transaction book.*
- *System Response: The system displays the selected book information and allows the user to modify it.*
- ***Stimulus:*** *User submits the updated book data.*
- *System Response: The system saves the changes and displays the updated transaction book information.*
- ***Stimulus:*** *User selects Delete Book on an existing transaction book.*
- *System Response: The system asks for confirmation before removing the selected transaction book.*
- ***Stimulus:*** *User confirms the deletion.*
- *System Response: The system deletes the selected transaction book and removes it from the displayed book list.*
- ***Stimulus:*** *User selects one transaction book from the available list.*
- *System Response: The system marks the selected transaction book as active and uses it for related financial records and transactions.*

4.2.3 Functional Requirements

- **REQ-4.2.1:** The system shall allow authenticated *Users* to create a new transaction book.
- **REQ-4.2.2:** The system shall require a book name when creating a transaction book.
- **REQ-4.2.3:** The system shall display all transaction books that belong to the authenticated *User*.
- **REQ-4.2.4:** The system shall allow *Users* to edit an existing transaction book.
- **REQ-4.2.5:** The system shall allow *Users* to delete an existing transaction book.
- **REQ-4.2.6:** The system shall request confirmation before permanently deleting a transaction book.
- **REQ-4.2.7:** The system shall allow *Users* to select one transaction book as the active book.
- **REQ-4.2.8:** The system shall ensure that transaction books are only accessible by their respective owners.

4.3 Add Transaction

4.3.1 Description and Priority

Priority: High. This feature allows authenticated Users to record financial transactions in the system. It is one of the main functions of the Money Manager System because users need it to track income and expenses, assign categories, and add notes for better financial management.

4.3.2 Stimulus/Response Sequences

- **Stimulus:** An authenticated *User* opens the Add Transaction feature.
- System Response: The system displays a transaction form for entering transaction details.
- **Stimulus:** *User* chooses to record an income transaction.
- System Response: The system allows the user to enter the income amount and related transaction information.
- **Stimulus:** *User* chooses to record an expense transaction.
- System Response: The system allows the user to enter the expense amount and related transaction information.
- **Stimulus:** *User* assigns a category to the transaction.
- System Response: The system records the selected category as part of the transaction data.
- **Stimulus:** *User* adds a note to the transaction.
- System Response: The system stores the note together with the transaction record.
- **Stimulus:** *User* submits the transaction form with valid data.
- System Response: The system saves the transaction and updates the related financial records.
- **Stimulus:** *User* submits incomplete or invalid transaction data.
- System Response: The system displays an error message and asks the user to correct the input before saving.

4.3.3 Functional Requirements

- **REQ-4.3.1:** The system shall allow authenticated *Users* to add a new transaction.
- **REQ-4.3.2:** The system shall allow *Users* to record a transaction as either income or expense.
- **REQ-4.3.3:** The system shall require the transaction amount to be entered before saving.
- **REQ-4.3.4:** The system shall allow *Users* to assign a category to a transaction.
- **REQ-4.3.5:** The system shall allow *Users* to add a note to a transaction.
- **REQ-4.3.6:** The system shall associate each transaction with the active transaction book.
- **REQ-4.3.7:** The system shall save the transaction after valid data is submitted.

- **REQ-4.3.8:** The system shall display an error message if the transaction data is incomplete or invalid.

4.4 View Transaction

4.4.1 Description and Priority

Priority: High. This feature allows authenticated Users to view recorded financial transactions. It helps users review their financial activities and monitor income and expense records that have been stored in the system.

4.4.2 Stimulus/Response Sequences

- **Stimulus:** An authenticated *User* opens the View Transaction feature.
- System Response: The system displays a list of recorded transactions from the selected transaction book.
- **Stimulus:** *User* selects a transaction from the list.
- System Response: The system displays the details of the selected transaction.
- **Stimulus:** *User* views transaction records.
- System Response: The system shows transaction information such as type, amount, category, and note.

4.4.3 Functional Requirements

- **REQ-4.4.1:** The system shall allow authenticated *Users* to view recorded transactions.
- **REQ-4.4.2:** The system shall display transactions based on the selected transaction book.
- **REQ-4.4.3:** The system shall display the type of transaction, whether income or expense.
- **REQ-4.4.4:** The system shall display the amount of each transaction.
- **REQ-4.4.5:** The system shall display the assigned category of a transaction.
- **REQ-4.4.6:** The system shall display the note associated with a transaction, if available.
- **REQ-4.4.7:** The system shall ensure that only the owner can view their transaction records.

4.5 View Wallet Balance

4.5.1 Description and Priority

Priority: Medium. This feature allows authenticated Users to view their current wallet balance. It helps users understand their financial condition by showing the balance calculated from recorded income and expense transactions.

4.5.2 Stimulus/Response Sequences

- **Stimulus:** An authenticated *User* opens the View Wallet Balance feature.

- System Response**:** The system calculates and displays the current wallet balance based on the stored transaction data.
- **Stimulus:** *User* adds a new income or expense transaction.
- System Response: The system updates the wallet balance according to the new transaction data.
- **Stimulus:** *User* selects a different transaction book.
- System Response: The system displays the wallet balance for the selected transaction book.

4.5.3 Functional Requirements

- **REQ-4.5.1:** The system shall allow authenticated *Users* to view their wallet balance.
- **REQ-4.5.2:** The system shall calculate the wallet balance based on recorded income and expense transactions.
- **REQ-4.5.3:** The system shall update the wallet balance whenever transaction data changes.
- **REQ-4.5.4:** The system shall display wallet balance information based on the selected transaction book.
- **REQ-4.5.5:** The system shall ensure that wallet balance data is only accessible by its owner.

4.6 View Charts / Analytics

4.6.1 Description and Priority

Priority: Medium. This feature allows authenticated Users to view simple charts and analytics of their financial data. It helps users understand spending patterns and financial activity in a more visual way.

4.6.2 Stimulus/Response Sequences

- **Stimulus:** An authenticated *User* opens the View Charts / Analytics feature.
- System Response: The system displays charts and summaries based on the user's recorded financial transactions.
- **Stimulus:** *User* views the analytics page.
- System Response: The system shows visual summaries such as expense distribution and transaction trends.
- **Stimulus:** *User* changes the selected transaction book.
- System Response: The system refreshes the charts and analytics based on the selected book's data.

4.6.3 Functional Requirements

- **REQ-4.6.1:** The system shall allow authenticated *Users* to view financial charts and analytics.
- **REQ-4.6.2:** The system shall display charts based on recorded transaction data.
- **REQ-4.6.3:** The system shall present simple summaries of income and expense activities.

- **REQ-4.6.4:** The system shall update chart data according to the selected transaction book.
- **REQ-4.6.5:** The system shall ensure that analytics data is only accessible by its owner.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

<If there are performance requirements for the product under various circumstances, state them here and explain their rationale, to help the developers understand the intent and make suitable design choices. Specify the timing relationships for real time systems. Make such requirements as specific as possible. You may need to state performance requirements for individual functional requirements or features.>

PERF-1: The page load time when navigating between main tabs (Books, Wallet, Charts) must not exceed 2 seconds on a standard internet connection (4G/5G).
PERF-2: Graphs on the "Charts" page must be rendered asynchronously to avoid blocking the user interface while fetching data from the database.

5.2 Safety Requirements

<Specify those requirements that are concerned with possible loss, damage, or harm that could result from the use of the product. Define any safeguards or actions that must be taken, as well as actions that must be prevented. Refer to any external policies or regulations that state safety issues that affect the product's design or use. Define any safety certifications that must be satisfied.>

SAFE-1: The system must prevent the accidental deletion of a *Transaction Book* or *Account* by displaying a secondary confirmation prompt (warning modal) before the deletion action is executed.

5.3 Security Requirements

<Specify any requirements regarding security or privacy issues surrounding use of the product or protection of the data used or created by the product. Define any user identity authentication requirements. Refer to any external policies or regulations containing security issues that affect the product. Define any security or privacy certifications that must be satisfied.>

SEC-1: All user passwords must be securely hashed using an algorithm such as bcrypt before being stored in the database.
SEC-2: The system must utilize the HTTPS protocol to encrypt data transmission between the client and the server.
SEC-3: Session authentication must employ token-based security (e.g., JSON Web Tokens / JWT) with an enforced session expiration time (e.g., timeout after 24 hours of inactivity).

5.4 Software Quality Attributes

<Specify any additional quality characteristics for the product that will be important to either the customers or the developers. Some to consider are: adaptability, availability, correctness, flexibility, interoperability, maintainability, portability, reliability, reusability, robustness, testability, and usability. Write these to be specific, quantitative, and verifiable when possible. At the least, clarify the relative preferences for various attributes, such as ease of use over ease of learning.>

Usability: The interface must prioritize a responsive mobile-first design (matching the Figma specifications), ensuring that buttons and links are easily tappable on mobile touch screens.
Reliability: The system must be capable of handling large transaction amounts (e.g., Rupiah formatting) without causing integer overflow errors within the database.

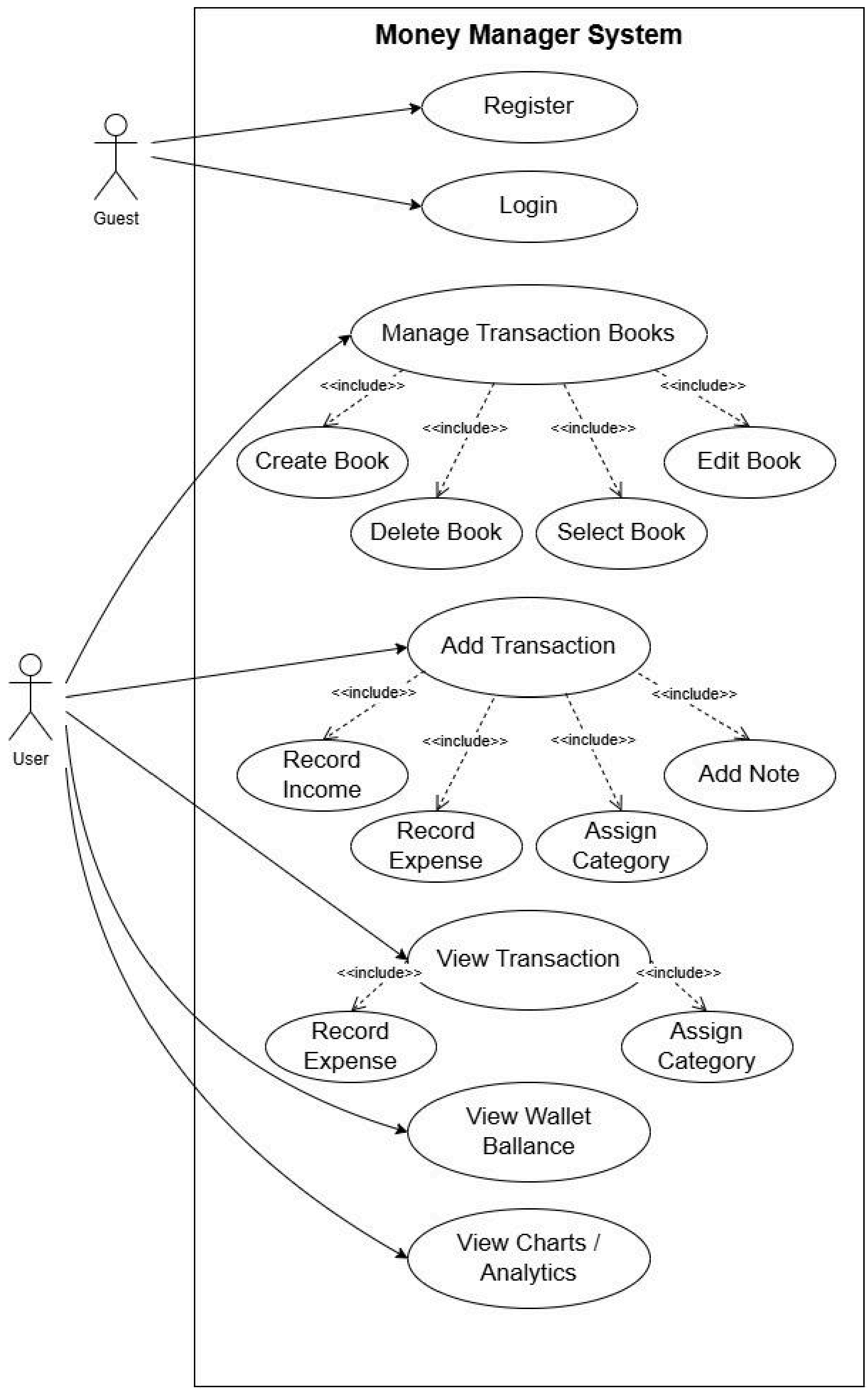
5.5 Business Rules

<List any operating principles about the product, such as which individuals or roles can perform which functions under specific circumstances. These are not functional requirements in themselves, but they may imply certain functional requirements to enforce the rules.>

BR-1: The system shall use the Rupiah (Rp) as the primary currency format for all displayed monetary values.
BR-2: The *Net Worth* value cannot be manually manipulated; its value is strictly a calculated sum derived from the recorded transactions and existing accounts.
BR-3: Data Isolation: Users may only view, add, modify, or delete transaction data and books that are directly associated with their own User ID.

6. Unified Modeling Languages

6.1 Use Case Diagram



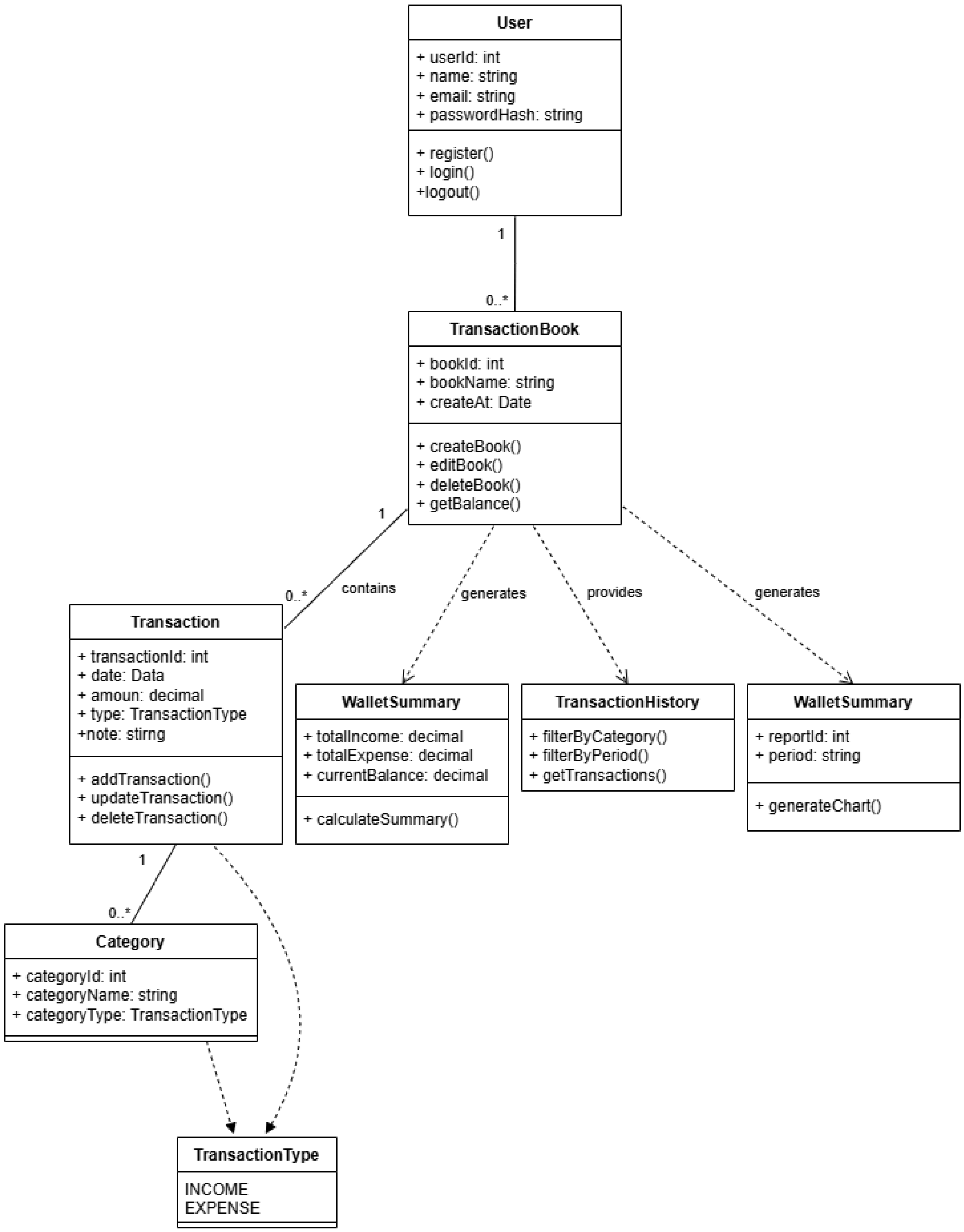
6.2 Activity Diagram

https://drive.google.com/drive/folders/1ei3rFPjWmAW5q8Q7VmmMGoWgd-k_X2wW?usp=sharing

6.3 Sequence Diagram

https://drive.google.com/drive/folders/1ei3rFPjWmAW5q8Q7VmmMGoWgd-k_X2wW?usp=sharing

6.4 Class Diagram



7. Prototype

7.1 Link Figma : Money Manager system – Figma

8. Other Requirement

<Define any other requirements not covered elsewhere in the SRS. This might include database requirements, internationalization requirements, legal requirements, reuse objectives for the project, and so on. Add any new sections that are pertinent to the project.>

Appendix A: Glossary

<Define all the terms necessary to properly interpret the SRS, including acronyms and abbreviations. You may wish to build a separate glossary that spans multiple projects or the entire organization, and just include terms specific to a single project in each SRS.>

Appendix B: Analysis Models

<Optionally, include any pertinent analysis models, such as data flow diagrams, class diagrams, state-transition diagrams, or entity-relationship diagrams.>

Appendix C: To Be Determined List

<Collect a numbered list of the TBD (to be determined) references that remain in the SRS so they can be tracked to closure.>